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Explainable and evidence-linked recommendations for spine surgery via a retrieval-augmented LLM agent

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ABSTRACT

Background: Clinical decision-making in spinal and spinal cord diseases requires a comprehensive assessment of imaging findings, neurological status, bone integrity, and patient-centered goals. Recently, the emergence of Large Language Models (LLMs) has provided new tools for intelligent decision support; however, their reliability and clinical interpretability remain to be systematically evaluated.

Methods: We propose a novel retrieval augmented generation (RAG)-based framework specifically tailored for spinal disease decision support and systematically evaluated four advanced LLM agents (Gemini 2.5 Flash, DeepSeek, GPT-4o, GPT-4o-mini). The framework integrates domain-specific prompting, structured response formatting, and evidence citation tracking.We used 200 real-world spinal cases, each involving diagnostic, therapeutic, and follow-up tasks. Five spine surgeons independently evaluated model outputs using an 11-dimension rubric; each dimension rated on a 5-point Likert scale to evaluate both clinical and technical performance. Inter-group differences were analyzed using the Kruskal–Wallis and Dunn's tests ($P < 0.05$), with radar plots used for multidimensional visualization.

Results: The proposed expert-evaluated framework enables a comprehensive, real-case-based comparison of four LLM agents. Gemini 2.5 Flash achieved the highest overall score (49.25 ± 2.88), significantly outperforming DeepSeek (47.49 ± 3.34, $P < 0.001$), GPT-4o (45.59 ± 3.89, $P < 0.001$), and GPT-4o-mini (41.23 ± 5.96, $P < 0.001$). It demonstrated leading performance particularly in humanistic care, follow-up suggestion, and test recommendation. DeepSeek showed superior capability in differential completeness (mean = 4.76), significantly outperforming the other three models ($P < 0.001$). Although GPT-4o-mini demonstrated stable system performance (mean = 4.61), it underperformed in core clinical reasoning dimensions. These findings reveal substantial inter-model variability in spine-specific clinical reasoning, an aspect often overlooked in prior non-benchmark LLM evaluations.

Conclusion: Among the four evaluated LLM agents, Gemini 2.5 Flash and DeepSeek demonstrated superior clinical accuracy, comprehensiveness, and usability in spine-related decision support. These findings support the potential of domain-adapted RAG agents to enhance evidence-based spinal care by providing accurate and comprehensive decision support. Future research should focus on multimodal integration data (e.g., imaging and clinical notes) and conducting prospective validation in real-world clinical environments.

1. Introduction

Spine-related disorders have received increasing attention globally, and according to the latest GBD 2021 data, the global prevalence of neck pain is 588.53/100,000, corresponding to approximately 43.286 million cases, while the prevalence of lower back pain reaches 3381.84/100,000, with more than 267 million cases [1]. The prevention and treatment of spinal disorders, one of the most disabling health problems in the world, is facing serious challenges.

Clinical decision-making for spinal diseases such as tumors, infections, congenital malformations, and trauma is a highly complex process that balances multiple dimensions of information such as

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0720-048X/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

imaging characteristics, neurological status, bone metabolism indicators, and patient expectations, which is reflected in the following aspects: Imaging is the important test that can be used by a clinical patient to determine the presence of a disease. Study showed that 30% of MRIs in the asymptomatic population showed disc herniation, 60% showed disc degeneration (age $\geq$ 40 years), and 80% of people over 50 years of age had imaging manifestations of degeneration without corresponding symptoms [2]. However, the correlation between imaging data and clinical symptoms may be poor at times [3]. Meanwhile, relying on symptoms or a single imaging test as a single criterion for clinical diagnosis is clearly unreliable. At the same time, for patients with abnormal nerve function, active treatment can have better efficacy, a study of the American Medical Association showed that among patients with neurogenic symptoms, compared with the non-surgical care group, the surgical group had a more significant functional improvement (ODI improvement $\geq$ 30%) at 1-year follow-up [4]. It is easy to see that the assessment of the deficit status of neurological function during treatment is also crucial. In addition to the above, bone status is also an important reference for making clinical decisions; bone status can affect patient treatment and recovery [5,6], and patients with poor bone status may be more prone to complications during the treatment process. In addition to objective examination or status, the patient's own psychological factors are particularly important, and several studies have shown that psychological factors are also capable of influencing the decision-making process for treatment [7], and are even directly related to prognosis [8]. In summary, spinal clinical decision-making must synthesize multiple domains of information—objective findings (imaging, neuro-status, bone health) and subjective elements (psychological readiness and personal preferences)—to ensure balanced, evidence-based treatment pathways.

Clinical decision making in spine surgery is a two-fold process: on the one hand, there is a need for accurate application of guidelines (e.g., NASS lumbar spine guidelines), and on the other hand, there is a need to incorporate “experience-driven” decision making methods. However, we cannot avoid some phenomena in this decision-making process: for example, there is a significant discrepancy between guideline recommendations and clinical practice – Lurie et al. pointed out in a multicenter study that only 58% of lumbar spine surgical decisions were fully compliant with NASS guidelines [9] (e.g., the 6-week trial period of conservative treatment), Due to cognitive biases, there is a potential risk of driving decision-making only through experience [10]. The limitations of this “guideline-experience” binary decision-making model highlight the urgent need for new decision support tools. The aim of this article is to explore the potential of AI to participate in clinical decision making.

And in this complex process, the use of decision aids can improve physician efficiency, and they may play a more active role in decision making and more accurate risk perception [11]. Therefore, in the current environment of rapid development of big models and artificial intelligence [12–15], our team expects that there will be mature AI products to effectively improve the ecology of clinical decision making, and to help clinical decision making as efficient decision aids for clinicians.

Advanced AI products such as GPT and DeepSeek [16] have demonstrated powerful medical data analytics [17], many studies have examined the performance of different large models in various domains: nursing [18], medical Image Generation [19,20], oncology [21,22], endocrine [23], etc. For spinal surgery, in a trial of ChatGPT intervening in cauda equina syndrome it was shown that ChatGPT may be a valuable aid in high-risk clinical decision-making, but studies have further demonstrated the need for the tool's validation and improvement [24]. However, the core issues that preclude direct clinical use are Hallucination [25,26] and Non-traceability [27,28]. In Sallam M's study, the GPT-4 was found to have a high rate of hallucination of 28.3% in open domain medical QA, and it is worth noting that the hallucination phenomenon was more pronounced in non-English speaking scenarios. phenomenon is more significant in non-English language in the scenarios (about 12% increase in error rate) [29]. In addition the phenomenon of inference paths not being able to be reasoned retrospectively when GPT-4 generates a differential diagnosis occurs in 64% of cases, and the product diagnosis is 19% less correct compared to the physician's diagnosis [30]. In summary, despite their potential, conventional AI decision-making tools remain limited in clinical application due to inherent shortcomings.

This study developed a clinical decision support system for spinal and spinal cord diseases based on Retrieval-Augmented Generation (RAG) technology [31], aiming to provide scientific and personalized diagnostic and treatment decision support for clinicians. Retrieval enhancements can increase the evidential support of generated content from 64% to 89% [32], making the system different from conventional AI products. The system adopts a modular architecture, integrating cutting-edge Large Language Models (LLMs) with professional medical knowledge bases, to provide evidence-based medical support for various types of spinal and spinal cord diseases through intelligent retrieval and reasoning mechanisms.

The system design adheres to three core principles: (1) clinical practice-oriented, ensuring that system outputs comply with evidence-based medical principles; (2) emphasis on personalized treatment strategies, providing customized recommendations based on patients' specific physiological characteristics (such as age, affected segments, pathological type) and imaging features; (3) maintaining decision transparency, clearly annotating recommendation basis and reference sources. The system innovatively implements a knowledge source citation visualization mechanism, allowing clinicians to directly view the specific knowledge sources supporting each diagnostic and treatment recommendation, enhancing the credibility and traceability of system outputs. This design philosophy permeates all levels of the system architecture, ensuring both clinical applicability and meeting medical ethical requirements. After the development of the system was completed, we validated its clinical utility and evaluated the potential of the RAG intelligences for clinical application by blind scoring of 200 cases by five specialists. This study aims to explore the feasibility of applying large language modeling in spine surgery and provides an important methodological reference for the development of smart healthcare.

2. Materials and methods

2.1. Study design and Ethics

2.1.1. Study design

This was a single-center, retrospective study designed to develop and validate an artificial intelligence (AI) model. The study adhered to the STARD [33] guidelines to ensure transparent reporting of model development, validation, and performance evaluation.

2.1.2. Ethics

Ethical approval for this study was waived by the Ethics Committee of The Third Affiliated Hospital of Southern Medical University because this study involved only retrospective analysis of de-identified clinical data. All procedures were in accordance with the Declaration of Helsinki.

2.2. System development

2.2.1. Overall architecture

The “Spinal Cord Disease Clinical Decision Support Intelligence” system developed in this study is based on a modular microservice architecture, integrating components such as front-end interactive interface, back-end RAG engine, knowledge base management and LLM modeling service, supporting structured and unstructured data input in the clinical process of spinal surgery, and generating diagnostic and

treatment recommendations with traceable literature and references automatically. automatically generate diagnosis and treatment recommendations with traceable literature references (Fig. 1).

The system consists of five interoperable core modules (Fig. 2). The front-end interface module has a conversational design that allows the clinician to enter detailed information about the patient's characteristics (e.g., "58-year-old female, progressive numbness and clumsiness of the hands for 6 months, unsteady gait for 2 months; MRI shows multisegmental cervical stenosis at C3-C6, with altered spinal cord signals at the C4/5 level"), and to present, in a structured format, the generated diagnosis and treatment recommendations. The data flow logic is emphasized: user input → query processing → retrieval recall → cue word splicing → model generation → recommendation output and literature visualization.

The system is containerized and deployed based on Docker and Kubernetes deployed in an A100 × 2 GPU cluster to satisfy multi-doctor concurrent calls, and supports MongoDB (database storage information and logging), Milvus(vector index) and PostgreSQL(relational structured backup), The three types of databases constitute a hybrid persistence architecture.

2.2.2. *Knowledge base Curation*

In this study, the medical knowledge base was constructed through a systematic strategy, and the inclusion criteria and data cleaning were as follows: the explicit inclusion criteria were: (1) published by authoritative organizations, (2) after 2017, and (3) containing diagnostic and treatment process or outcome data; the exclusion items included duplicated versions, summary-type secondary literature, and lack of full-text literature. All documents are standardized and cleaned, format converted, and redundant symbols and header and footer interference are removed.

The system has a built-in timed update mechanism that runs a Cron Job every six months to compare new versions of literature through PubMed and RSS feeds, triggering an incremental update of the vector library and generating new version tags to ensure that citations are always kept up-to-date. Fig. 3 describes the five-step process of the knowledge base, from initial collection, standard format conversion, expert review, accessioning to the database, and version tracking.

2.2.3. *Embedding & vector Store*

To ensure semantic coherence and domain specificity, this study implemented an optimized segmentation strategy tailored to spinal cord disease–related texts. All documents in the knowledge base were processed using a hybrid segmentation algorithm that limited each paragraph to approximately 300–500 characters. Special attention was given to preserving spine-specific terminology—such as "OPLL" and "atlantoaxial dislocation"—within the same segment to maintain contextual integrity and enhance semantic recall. This approach was designed to improve the relevance and accuracy of downstream retrieval and inference tasks.

A key determinant of retrieval system performance is vector embedding. After evaluating multiple embedding models, the text-embedding-3-large model provided by OpenAI (cited in the literature) is used to generate a vector representation of the literature, which is experimentally superior to text-embedding-ada-002 and the homegrown model, and is more advantageous in handling medical abbreviations and synonyms (e.g., " ASIA D" and" incomplete spinal cord injury").

Vector Retrieval System and Performance Parameters:Literature vectors are indexed by IVF_FLAT constructed by Milvuswith index parameters set to nlist = 4096, nprobe = 16.The system achieves 92.4% accuracy of Top-4 text recall in A100 dual-card environment, and the P95 latency is controlled to be within 23 ms, which supports the demand of real-time response.

2.2.4. *Prompt engineering*

Prompt engineering serves as the core link in transforming the general capabilities of large language models (LLMs) into specialized clinical decision support for spinal and spinal cord diseases. In response to the reviewer's suggestion to provide explicit prompts and templates, this study designed a systematic, domain-adapted prompt system integrating role definition, task constraints, structured templates, and few-shot learning. This ensures model outputs are evidence-based, standardized, and aligned with spinal surgical clinical workflows.

This system prompt word (prompt) by the Role (role settings), Task (task description), Context (knowledge references), Template (output format) four parts, embedded in the system prompt to unify the guidance model output specification of structured answers:

Role settings: Explicitly designates the model as a "specialized

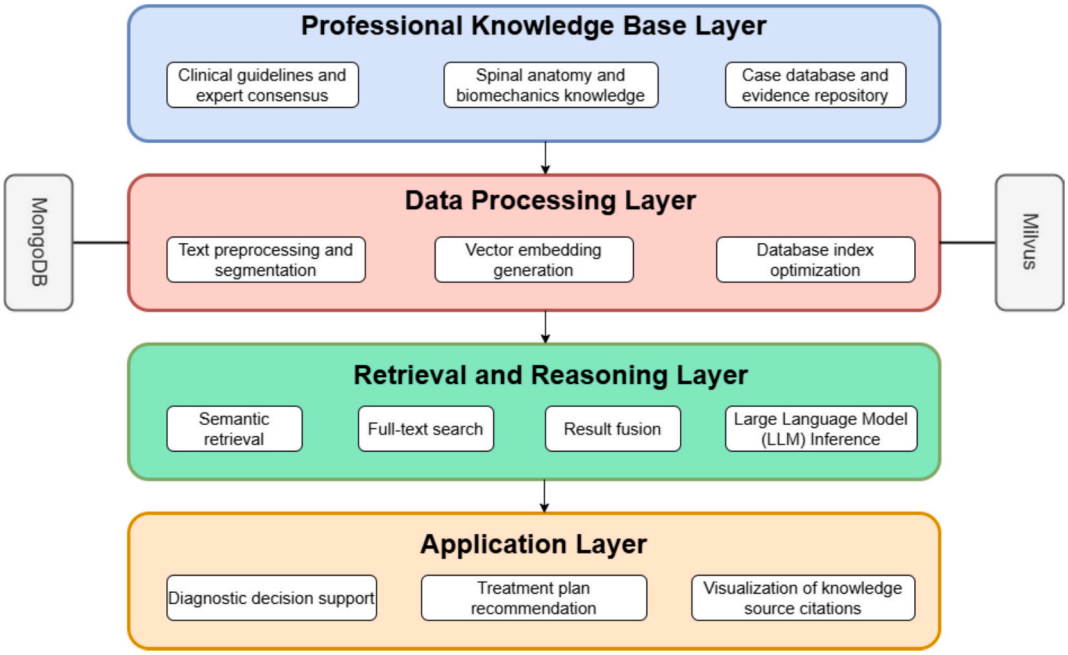


Fig. 1. System architecture diagram of the clinical decision support system for spinal and spinal cord diseases.

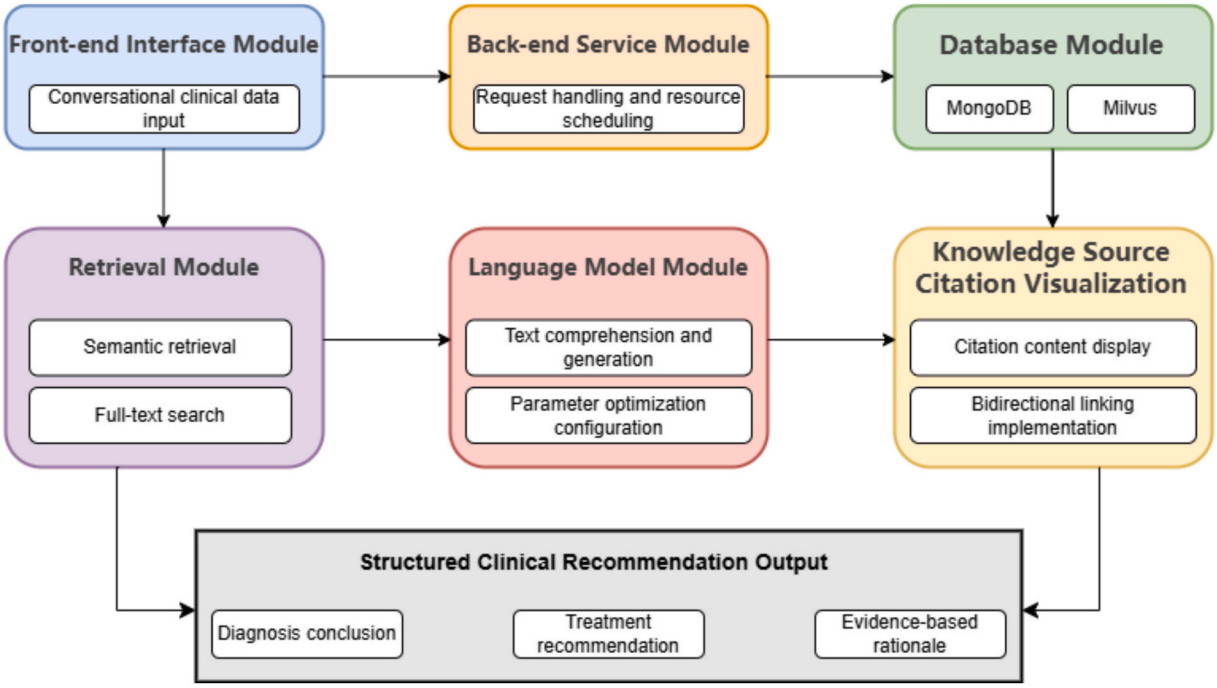


Fig. 2. System module composition and data flow diagram.

clinical decision support assistant for spinal and spinal cord diseases," confining responses to spinal surgery–relevant diagnosis, treatment, and follow-up to avoid off-domain outputs.

Task Description: Mandates the model to generate structured, evidence-linked recommendations based on the top-4 retrieved literature paragraphs and user-provided clinical information, prioritizing patient-centric concerns (e.g., disease severity, surgical necessity).

Knowledge Reference: Dynamically embeds the top-4 most relevant literature segments from the knowledge base, each tagged with a unique literature ID, source (e.g., NASS guidelines), publication year, and evidence level (e.g., Level 1A) to enable traceability.

Output Format: Prescribes a standardized response framework and formatting rules to ensure consistency across different cases and models.

Contextual paragraph referencing and dynamic injection mechanism. The retrieved Top-4 paragraphs are automatically spliced into Context paragraphs, and carry unique literature ID and traceable path; user input information is uniformly embedded in User Prompt, which is spliced with the context of the knowledge paragraph and input into the large language model together.

This system builds a Few-shot [34] learning example library containing 50 typical cases covering different subtypes and severity levels of common spinal cord diseases. Each example contains complete clinical information, standard diagnostic procedures and expert treatment recommendations as a reference for model learning. The experiment adopts a dynamic example selection mechanism, which automatically selects the most relevant 3–5 examples based on the characteristics of the query case (e.g., age group, main symptoms, and imaging manifestations) as inputs for Few-shot learning. The system automatically matches historical cases with similar age, disease type, and symptoms to the current case, and adds template responses to the prompt to guide the model to produce a consistent wording style (e.g., "JOA score", "ASIA classification", and other specialized expressions).

The system incorporates a carefully engineered prompting strategy. Specifically, all system-generated responses are strictly constrained to reference only the content retrieved from the medical knowledge base. In scenarios where the available literature does not provide a definitive answer, the model is explicitly instructed to respond with standardized statements such as "no clear conclusions available," thereby avoiding speculative or unsupported claims. Furthermore, the output format adopts a structured "recommendation + cited paragraph" layout, wherein each diagnostic or therapeutic suggestion is directly linked to its corresponding evidence source. This design allows clinicians to view the precise literature fragment supporting each recommendation through clickable citations, enabling transparent traceability and facilitating deeper understanding of the medical rationale. By ensuring that each model output is anchored to verifiable evidence, the system enhances both the reliability and interpretability of its clinical decision support.

2.3. LLM-Agent evaluation cohort

2.3.1. Case selection

A total of 200 patient cases diagnosed within the past two years in our department were retrospectively collected. The inclusion criteria were designed to encompass a broad spectrum of spinal pathologies, including degenerative disc disease, spinal stenosis, spondylolisthesis, and spinal deformities. Each case was required to have comprehensive imaging data—MRI, CT, and X-rays—as well as detailed clinical documentation. Cases with incomplete records or involving conditions beyond the scope of standard spinal surgical interventions were excluded.

The baseline characteristics of the 200 included patients were as follows: The mean age was 54.3 ± 10.2 years (range: 11–74 years), with 89 male patients (44.5%) and 111 female patients (55.5%). In terms of age stratification, 1 patient (0.5%) was under 18 years old (diagnosed with adolescent idiopathic scoliosis), 35 patients (17.5%) were aged 18–44 years, 124 patients (62.0%) were aged 45–64 years, and 40 patients (20.0%) were aged 65 years or older.

In terms of disease distribution, degenerative spinal diseases were the main pathological types. Lumbar spinal stenosis was the most common diagnosis (76 cases, 38.0%), followed by mixed cervical spondylosis (28 cases, 14.0%), lumbar spondylolisthesis (22 cases, 11.0%), spinal fractures (20 cases, 10.0%, including 12 cases of osteoporotic pathological fractures in the elderly), discogenic low back pain/lumbar disc herniation (18 cases, 9.0%), spinal deformities (8 cases, 4.0%), intraspinal space-occupying lesions (4 cases, 2.0%), spinal

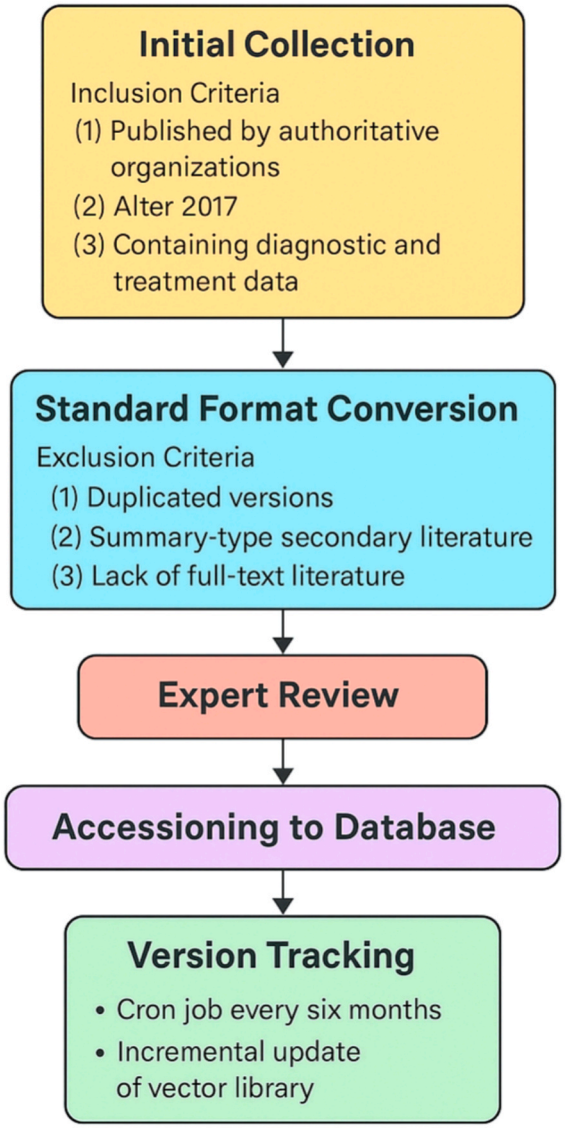


Fig. 3. Literature screening and standardization flowchart for medical knowledge base construction.

infections (3 cases, 1.5%), and other conditions (11 cases, 5.5%). Comorbidities were present in 63 patients (31.5%), with hypertension (38 cases, 19.0%) and type 2 diabetes mellitus (21 cases, 10.5%) being the most common. In terms of involved spinal segments, lumbar lesions accounted for 71.0% (142 cases), cervical lesions for 17.5% (35 cases), thoracic lesions for 6.5% (13 cases), and multi-segment combined lesions for 5.0% (10 cases).

2.3.2. Participating physicians

To ensure an objective evaluation, our study employed a blinded and randomized approach. Initially, researchers randomly selected 200 complete clinical cases from our department's database and input them into the four LLM agents we developed: gemini2.5flash (https://ai.host.aiforme.site/chat/share?shareId=wh2t7uya031uxvysg3zjei2a), deepseek (https://ai.host.aiforme.site/chat/share?shareId=s9ytu2byry2u9hjkef2j6h5e), GPT-4o (https://ai.host.aiforme.site/chat/share?shareId=nvyml796hupla8bd5b03erbg) and GPT-4o-mini (https://ai.host.aiforme.site/chat/share?shareId=en83lhtl76rprxbkkajzj1s3). The responses generated by these agents, including content and citation information, were converted into plain text to remove any identifiable model features and compiled into Excel spreadsheets without any subjective interpretation. Subsequently, five experienced spinal surgeons independently reviewed the responses in these spreadsheets according to a predefined scale across 11 dimensions, using a 5-point Likert scale for scoring. Finally, researchers aggregated the scores and performed statistical analyses to compare the professionalism and technical performance of the responses from the four LLM agents.

2.3.3. Scoring Instrument

Two groups of evaluation criteria were established to assess the performance of the LLM agents (Table 1). (1) Professionalism metrics included diagnostic accuracy, differential completeness, test recommendation, treatment appropriateness, complication alerts, follow-up Suggestion. (2) Technical performance metrics included humanistic care, follow-up response quality, search relevance, system stability, interaction friendliness. These criteria were adapted from validated frameworks in the literature assessing clinical accuracy and usability of LLM in healthcare settings [35,36].

2.4. Statistical analysis

All descriptive statistics and analyses were conducted using SPSS 25.0 (SPSS Inc., Chicago, IL) and R (Version 4.2.2, R Foundation, Vienna, Austria). For each evaluation dimension, the average scores provided by five independent raters were calculated and reported as mean ± SD, along with the corresponding median values and range (maximum–minimum) [37]. The Shapiro-Wilk test indicated that all comparative index data were non-normally distributed; therefore, the Kruskal-Wallis test was employed for inter-model comparisons. Statistical significance was defined as $P < 0.05$. When significant differences were observed, Dunn's post hoc test was used for pairwise comparisons,

Table 1
Summarizes the evaluation dimensions and scoring definitions.

Dimension	Metric Name	Scoring Definition
Professionalism	Diagnostic Accuracy	Correctness of the primary diagnosis based on the clinical presentation and available data.
Professionalism	Differential Completeness	Extent to which relevant differential diagnoses are thoroughly considered and appropriately ranked.
Professionalism	Test Recommendation	Appropriateness and completeness of recommended ancillary tests (e.g., MRI, lab work).
Professionalism	Treatment Appropriateness	Adequacy of warnings regarding potential complications and adverse outcomes.
Professionalism	Complication Alerts	Proactive identification of possible complications
Professionalism	Follow-up Suggestion	Completeness and clarity of follow-up care instructions, including monitoring and rehabilitation.
Technical Performance	Humanistic Care	Empathy, patient-centeredness, and use of supportive language in the response.
Technical Performance	Follow-up Response Quality	Relevance and completeness of answers to follow-up or clarification questions.
Technical Performance	Search Relevance	Accuracy and precision of cited evidence in support of recommendations
Technical Performance	System Stability	Consistency and reliability of the system across multiple interactions
Technical Performance	Interaction Friendliness	Clarity, readability, and overall ease of communication in the response.

Each dimension was rated on a 5-point Likert scale by five spine surgeons:
1 = Poor: completely incorrect or irrelevant.
2 = Fair: partially correct but with major flaws.
3 = Acceptable: generally reasonable but incomplete.
4 = Good: accurate and mostly complete.
5 = Excellent: fully correct, comprehensive, and contextually appropriate.

with *P*-values adjusted via Bonferroni correction. In addition, Kendall's W coefficient was calculated to assess inter-rater agreement and Cronbach's α coefficient was used to evaluate the internal consistency of the scoring scale, with $\alpha \geq 0.7$ considered indicative of acceptable reliability.

2.5. Implementation Details

We selected FastGPT (v0.9.3) as the foundational framework due to its robust support for knowledge base management, vector similarity retrieval, and modular AI workflow orchestration. FastGPT's extensible plugin system enabled seamless integration with domain-specific prompting mechanisms and evidence-grounded citation chains, facilitating traceable clinical reasoning pathways. The front-end interface was developed using Next.js (v13.4), TypeScript (v4.9), and Chakra UI (v2.5.1). This technology stack enabled responsive design, consistent theming, and a smooth interaction experience across platforms, particularly optimized for clinicians operating on varied screen sizes (e.g., desktops, tablets). The back-end employed a hybrid storage architecture. User credentials, dialogue history, and operational logs were stored in MongoDB (v6.0), ensuring flexible document-based schema support and high write throughput. Meanwhile, the embedding vectors—generated via a fine-tuned BGE-Large-En (v1.5) model—were stored in Milvus (v2.3.1). The system used the IVF_FLAT index type with cosine similarity as the metric to balance semantic matching accuracy and query efficiency in medical contexts. All modules were containerized and deployed using Docker (v24.0) with Kubernetes (v1.27) orchestration, enabling dynamic scaling and efficient resource management in cloud environments.

3. Results

3.1. Knowledge base characteristics

The knowledge base consists of 44 authoritative publications in the field of spinal surgery, including 8 clinical guidelines, 10 expert consensus statements, 6 textbook chapters, 12 systematic reviews, and 8 representative case series (Table 2). These sources cover common clinical scenarios such as spinal degenerative diseases, spinal tuberculosis, traumatic injuries, and spinal cord compression, and can provide evidence-based recommendations for preoperative, intraoperative, and postoperative management.

3.2. System Functionality Overview

The system integrates multiple functional modules based on LLMs, aiming to be consistent with real-world Diagnostic and therapeutic workflows. First, a conversational structured input interface allows clinicians to enter detailed patient information—including symptoms, imaging findings, and history—in free-text form, which is automatically parsed into structured data formats for further analysis. Second, the system employs a hybrid Retrieval-Augmented Generation (RAG) mechanism, which combines semantic vector retrieval with keyword-based full-text search, ensuring both conceptual relevance and lexical precision in sourcing evidence from the knowledge base. Third, a citation visualization module displays original knowledge sources—such as clinical guidelines, expert consensus statements, or textbook chapters—alongside the model's recommendations in an interactive format, thereby enhancing transparency and interpretability. Fourth, the system incorporates few-shot learning with dynamic prompt templates, utilizing a curated case library to adapt prompts in real time based on the clinical features of the query, improving task specificity and diagnostic accuracy. Finally, the system supports parameter-controlled generation, allowing adjustment of key values such as *temperature* and *top-p* to balance determinism and diversity in model output depending on the clinical scenario.

We have developed four clinical decision support systems for spinal and spinal cord diseases by integrating the system with foundational large models—Gemini 2.5 Flash, DeepSeek-R1, GPT-4o, and GPT-4o-mini. These systems have been deployed in a clinical testing environment, allowing physicians to input patient information and automatically generate tiered diagnoses, differential diagnoses, and treatment recommendations. Feedback from interactive testing with multiple spine surgeons indicates that these systems consistently produce highly professional and logically coherent multi-level medical recommendations, supplemented with literature citations. The outputs are characterized by excellent readability and clinical acceptance.

Table 2
Main components of the spinal and spinal cord disease knowledge base.

Type of Literature	Quantity	Content Coverage	Publication / Update Time
Clinical Guidelines	8	Diagnostic criteria, treatment protocols, prognostic evaluation	2019–2022
Expert Consensus Statements	10	Surgical indications, technical standards, rehabilitation management	2018–2022
Textbook Chapters	6	Anatomical foundations, biomechanics, pathophysiology	2017–2022
Systematic Reviews	12	Latest research progress, treatment controversies, new technology evaluation	2020–2022
Case Series	8	Typical/complex case management, surgical techniques, complication handling	2018–2022

3.3. Physician scoring outcomes

In the evaluation of 200 real-world cases, five spine surgeons independently assessed each intelligent agent using a standardized scoring scale, yielding a total of 44,000 individual scores (8800 per rater). The results showed that the gemini2.5-flash agent achieved the highest overall performance, with a mean total score of 49.25 ± 2.88 points, significantly outperforming DeepSeek (47.49 ± 3.34, $P < 0.001$), GPT-4o (45.59 ± 3.89, $P < 0.001$) and GPT-4o-mini (41.23 ± 5.96, $P < 0.001$) (Table 3).

The Kruskal–Wallis test revealed statistically significant differences in model performance across all evaluation dimensions ($P < 0.001$ for all). Further analysis using Dunn's post hoc test and visualization through the radar plot (Fig. 4) demonstrated that Gemini 2.5 Flash and DeepSeek showed comparable and superior scores in diagnostic accuracy, treatment appropriateness, complication alerts, and follow-up response quality, outperforming GPT-4o and GPT-4o-mini in these dimensions.

Specifically, Gemini 2.5 Flash demonstrated leading performance in test recommendation (mean = 4.54), follow-up suggestion (mean = 4.72), and humanistic care (mean = 4.66), highlighting its strengths in comprehensive and empathetic output. DeepSeek achieved the highest score in completeness of differential diagnosis (mean = 4.76), significantly outperforming the other three models ($P < 0.001$). Although GPT-4o-mini received the highest score in system stability (mean = 4.61), it had the lowest scores across several other evaluation dimensions.

Additionally, the Cronbach's α coefficient for the scale was 0.77, indicating acceptable internal consistency for the survey. The Kendall's tau correlation among raters was 0.853 ($p < 0.001$), suggesting substantial agreement regarding the quality and relevance of system-generated recommendations.

3.4. Representative case analysis and evidence Alignment Demonstration

To further elucidate the distinctions among the four LLM agents, we conducted a detailed, evidence-traceable, case-based comparison. A

Table 3
Performance of four LLM agents in spine-related tasks.

	Gemini2.5flash	Deepseek	GPT-4o	GPT-4o-mini	*P*-value*
Diagnostic Accuracy					
Mean ± SD	4.26 ± 0.8[abc]	4.38 ± 0.73[ab]	4.17 ± 0.66[ac]	3.68 ± 0.9[d]	<0.001
Median (min–max)	4.0(2.0–5.0)	5.0(2.0–5.0)	4.0(2.0–5.0)	4.0(1.0–5.0)	
Differential Completeness					
Mean ± SD	4.58 ± 0.49[a]	4.76 ± 0.72[b]	4.01 ± 0.79[b]	3.28 ± 0.79[c]	<0.001
Median (min–max)	5.0(4.0–5.0)	4.0(3.0–5.0)	4.0(2.0–5.0)	3.0(4.0–5.0)	
Test Recommendation					
Mean ± SD	4.54 ± 0.58[a]	4.18 ± 0.71[b]	3.99 ± 0.58[c]	3.39 ± 0.88[d]	<0.001
Median (min–max)	5.0(3.0–5.0)	4.0(2.0–5.0)	4.0(2.0–5.0)	3.0(1.0–5.0)	
Treatment Appropriateness					
Mean ± SD	4.49 ± 0.66[a]	4.51 ± 0.61[a]	4.09 ± 0.75[b]	3.52 ± 1.04[c]	<0.001
Median (min–max)	5.0(3.0–5.0)	5.0(3.0–5.0)	4.0(2.0–5.0)	4.0(1.0–5.0)	
Complication Alerts					
Mean ± SD	4.24 ± 0.75[a]	4.18 ± 0.66[a]	3.8 ± 0.74[b]	3.38 ± 0.79[c]	<0.001
Median (min–max)	4.0(3.0–5.0)	4.0(3.0–5.0)	4.0(2.0–5.0)	3.0(1.0–5.0)	
Follow-up Suggestion					
Mean ± SD	4.72 ± 0.58[a]	4.39 ± 0.63[b]	3.89 ± 0.72[c]	3.46 ± 0.9[d]	<0.001
Median (min–max)	5.0(3.0–5.0)	4.0(3.0–5.0)	4.0(2.0–5.0)	4.0(1.0–5.0)	
Humanistic Care					
Mean ± SD	4.66 ± 0.5[a]	3.49 ± 1.08[b]	3.65 ± 0.77[b]	3.54 ± 0.72[b]	<0.001
Median (min–max)	5.0(3.0–5.0)	3.0(1.0–5.0)	3.0(2.0–5.0)	3.0(2.0–5.0)	
Follow-up Response Quality Queries					
Mean ± SD	4.59 ± 0.52[a]	4.5 ± 0.58[a]	4.13 ± 0.62[b]	3.59 ± 0.92[c]	<0.001
Median (min–max)	5.0(3.0–5.0)	5.0(2.0–5.0)	4.0(3.0–5.0)	4.0(1.0–5.0)	
Search Relevance					
Mean ± SD	4.77 ± 0.47[a]	4.8 ± 0.51[a]	4.76 ± 0.53[a]	4.56 ± 0.95[b]	<0.001
Median (min–max)	5.0(3.0–5.0)	5.0(3.0–5.0)	5.0(2.0–5.0)	5.0(1.0–5.0)	
System Stability					
Mean ± SD	3.98 ± 1.24[a]	4.26 ± 0.95[a]	4.48 ± 0.83[b]	4.61 ± 0.84[c]	<0.001
Median (min–max)	5.0(1.0–5.0)	5.0(1.0–5.0)	5.0(1.0–5.0)	5.0(1.0–5.0)	
Interaction Friendliness					
Mean ± SD	4.43 ± 0.78[a]	4.66 ± 0.6[b]	4.63 ± 0.62[b]	4.33 ± 0.96[a]	<0.001
Median (min–max)	5.0(2.0–5.0)	5.0(2.0–5.0)	5.0(2.0–5.0)	5.0(2.0–5.0)	
Total Score					
Mean ± SD	49.25 ± 2.88[a]	47.49 ± 3.34[b]	45.59 ± 3.89[c]	41.23 ± 5.96[d]	<0.001
Median (min–max)	49.0(41.0–55.0)	48.0(35.0–55.0)	46.0(34.0–54.0)	43.0(24.0–52.0)	
Response Time					
Mean ± SD	24.98 ± 19.46[ab]	27.13 ± 24.14[ab]	19.7 ± 12.23[ab]	20.1 ± 14.33[bc]	<0.001
Median (min–max)	18.4(4.2–113.6)	17.5(4.8–128.6)	16.4(2.7–96.4)	15.5(2.2–100.4)	

*Kruskal-Wallis Test was used to determine significant differences between four LLMs.

representative clinical scenario was used, involving a 58-year-old female patient presenting with progressive cervical myelopathy symptoms—specifically, bilateral hand numbness and clumsiness for six months and gait instability for two months, accompanied by mild neck pain. Physical examination revealed positive bilateral Hoffmann's and Babinski's signs, lower limb hyperreflexia, and a modified Japanese Orthopaedic Association (mJOA) score of 13. MRI confirmed multilevel cervical spinal canal stenosis from C3 to C6 with altered spinal cord signal at the C4/5 level. This case constitutes a diagnostically and therapeutically moderate-complexity scenario, ideal for evaluating nuanced clinical reasoning.

When processing this case through our Retrieval-Augmented Generation (RAG) framework, each model generated outputs that were not only compared against the final clinical treatment plan but also meticulously analyzed for their linkage to specific evidence retrieved from our curated spinal surgery knowledge base. The distinct characteristics of each agent, as revealed by this dual-focus analysis, are summarized below and detailed in Table 4.

Gemini 2.5 Flash exhibited the most thorough and logically structured output, standing out for its detailed diagnostic interpretations, comprehensive differential diagnosis (including conditions like ALS and multiple sclerosis), and in-depth pathophysiological reasoning. This comprehensiveness was enabled by its superior evidence synthesis capability. The model retrieved and integrated multiple relevant

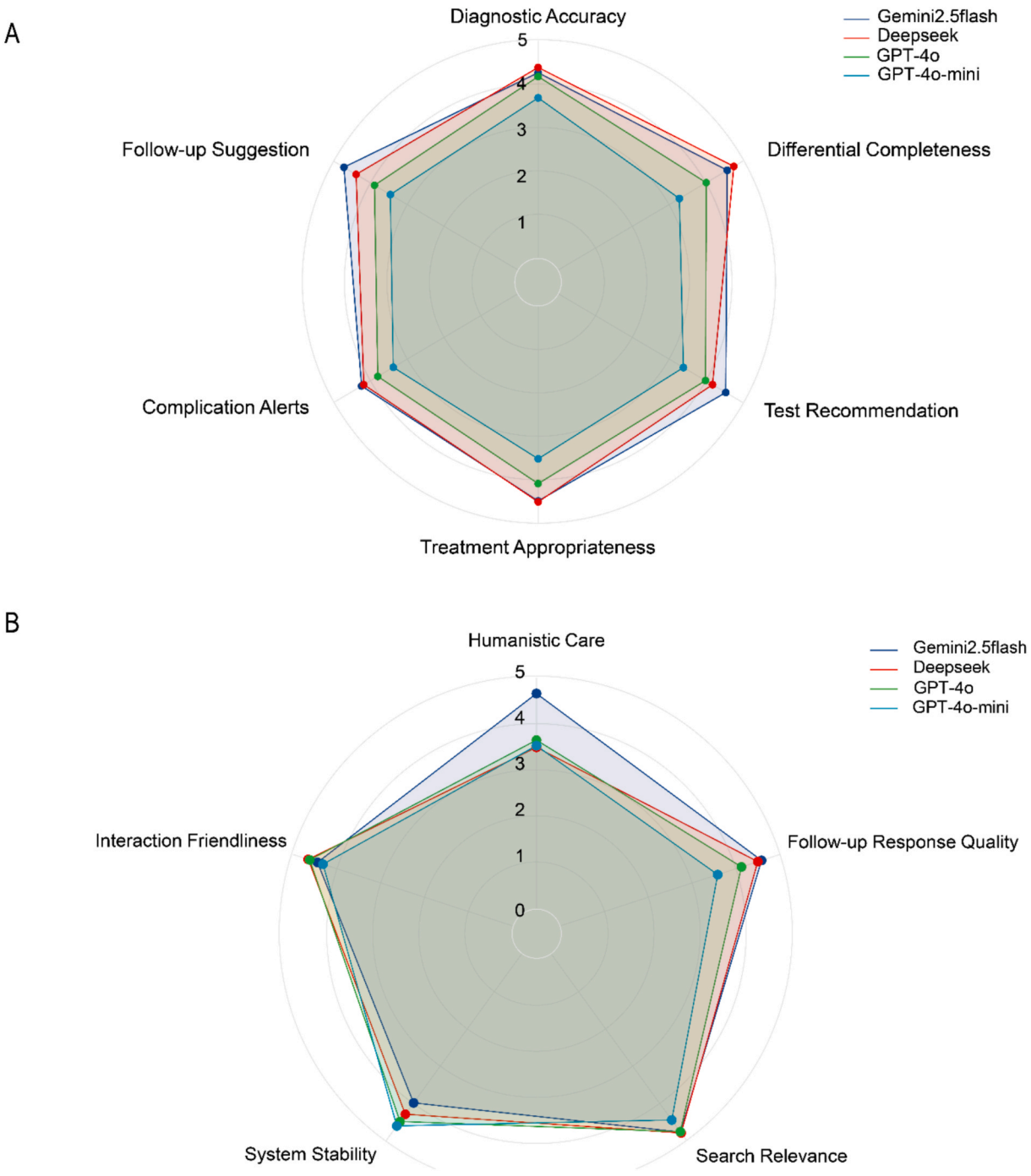


Fig. 4. Radar plot of four RAG-based LLM agents.(A) Radar plot of professionalism metrics. (B) Radar plot of technical performance metrics. Each axis represents the mean score across five spine surgeons using a 5-point Likert scale.

evidence fragments from the knowledge base, covering topics from MRI interpretation and multi-level stenosis management to the inappropriateness of certain procedures in unstable conditions. Although occasionally verbose, its content offered high educational value. Compared to DeepSeek, it was slightly less optimized for rapid, point-of-care decision-making but was better suited for deep case understanding, learning, and comprehensive preoperative review.

Deepseek, a domestically developed model, demonstrated strong applicability and high operational efficiency in real-world clinical settings. It produced concise, focused, and pragmatic diagnostic and therapeutic recommendations. Its performance was characterized by precise evidence anchoring. For instance, it verbatim cited a key segment from an authoritative textbook: Spinal cord signal can suggest inflammatory changes or gliosis to authoritatively explain the clinical significance of the MRI finding. This ability to ground critical diagnostic interpretations directly in sourced literature enhances the reliability and trustworthiness of its outputs for key clinical decisions, making it particularly suitable for frontline diagnostic confirmation and surgical counseling.

GPT-4o produced more comprehensive and logically structured outputs, featuring accurate diagnostic formulations and treatment plans supported by clear reasoning. A key strength was its explicit evidence grounding. The model specifically cited pathological reflexes (Babinski/Hoffmann signs) as indicators of chronic spinal cord injury from the literature and, most notably, explicitly referenced the "NASS

Table 4
Evidence-based comparison of LLM agent outputs for a representative cervical myelopathy case.

Agent	Diagnostic Approach & Key Suggestions	Examples of Retrieved & Cited Evidence	Evidence Alignment & Clinical Utility Profile
Gemini 2.5 Flash	Comprehensive diagnosis with extensive differentials; detailed discussion of surgical options and rehab.	Multiple evidence snippets on: MRI use, multi-level stenosis management, procedure appropriateness.	Deep Evidence Synthesis. Integrates broad evidence for comprehensive, educational output. Best for in-depth case analysis and trainee education.
DeepSeek	Accurate diagnosis with focused pathophysiological explanation of MRI signal change; clear surgical recommendation.	Verbatim textbook quote: Spinal cord signal can suggest inflammatory changes or gliosis to explain MRI finding.	Precise Evidence Anchoring. Excels at grounding specific diagnostic conclusions in sourced literature. Optimal for efficient, reliable diagnostic confirmation.
GPT-4o	Precise diagnosis of multilevel CSM; strongly recommended surgery (e.g., ACDF), citing guideline grade.	1. Literature on Babinski/ Hoffmann signs as signs of chronic cord injury.2. Direct reference to "NASS guideline" for surgical recommendation.	High Evidence-Based Alignment. Demonstrates a clear, auditable chain from findings to guideline-backed recommendations. Ideal for structured preoperative decision support.
GPT-4o-mini	Identified spinal cord compression; recommended surgical consultation and basic pre-op assessments.	General textbook descriptions linking symptoms (e.g., "gait instability") to spinal cord pathology.	Basic Alignment. Provides a safe, guideline-concordant foundation suitable for initial triage and patient communication. Lacks evidence for complex surgical planning.

guideline" as the basis for its Grade A recommendation for surgical intervention. This direct linkage to a recognized clinical guideline demonstrates a high level of evidence transparency. The content demonstrated moderate to high levels of evidence-based clinical reasoning and domain professionalism, making it highly suitable for preoperative planning support.

GPT-4o-mini generated a basic clinical analysis framework that successfully identified major symptoms and offered general treatment suggestions. Its response correctly pointed towards spinal cord compression and suggested surgical consultation, citing textbook-level evidence that links symptoms like gait instability to myelopathy. While the response structure was complete and suitable for rapid initial triage, the overall clinical depth and specificity were limited. It did not retrieve or cite more granular evidence regarding specific surgical indications, techniques, or high-grade guidelines, reflecting its role as a foundational screening tool.

4. Discussion

In this study, we designed and implemented four Retrieval-Augmented Generation (RAG)–based intelligent agents tailored for clinical decision support in spinal and spinal cord disorders. By integrating domain-specific knowledge bases with large language model inference, the agents were capable of generating context-aware, evidence-grounded recommendations across diagnosis, examination, and treatment dimensions. Comprehensive case-based evaluation and multi-dimensional analysis of clinical cases revealed clear performance differences among the four RAG agents, with several demonstrating notable strengths in output clarity, clinical relevance, and citation traceability. Notably, the structured "recommendation + evidence" format enabled transparent reasoning, supporting both interpretability and trustworthiness in clinical settings.

The RAG-based spinal intelligence agents offer significant advantages in clinical decision support for spinal surgery. First, the integration of specialized domain knowledge bases with advanced LLMs, combined with few-shot learning and dynamically adapted prompt templates, enhances the specificity and accuracy of diagnostic outputs in response to diverse clinical presentations. Second, the systems use a hybrid RAG mechanism combining semantic vector retrieval with keyword-based full-text search, allowing the model to generate well-grounded responses in complex spinal diagnostic and treatment scenarios. This improves both conceptual relevance and lexical precision while also mitigating the hallucination risks commonly associated with pure generative LLMs [38,39]. Third, the inclusion of an interactive citation visualization module allows clinicians to directly view the specific knowledge sources supporting each diagnostic and treatment recommendation. Furthermore, the systems employ a conversational structured-input interface and parses free-text patient descriptions into structured formats.

Recent studies have increasingly explored the application of LLMs in various medical domains, highlighting their potential and performance variability across different clinical tasks [40–42]. For example, Gültekin et al. found that both ChatGPT-4o and DeepSeek R1 delivered high-quality and accurate information for patient education on anterior cruciate ligament surgery, with ChatGPT-4o demonstrating greater comprehensiveness and DeepSeek R1 excelling in clarity and readability [43]. Similarly, Hassan et al. investigated the performance of ChatGPT-3.5, ChatGPT-4o, and Google Gemini in answering temporomandibular joint disorder–related queries. They reported that Gemini achieved the highest diagnostic accuracy, ChatGPT-4o provided the most complete responses, and both included citations and safety disclaimers. ChatGPT-3.5 exhibited the highest readability but lacked references [44]. The variability in evaluation outcomes observed across these LLMs may stem from differences in model architecture, training data composition, feedback mechanisms, and update cycles, all of which influence how each model handles clinical accuracy, response completeness, and safety considerations. Among the four RAG-based agents evaluated, Gemini 2.5 Flash and DeepSeek consistently outperformed the other models across the majority of evaluation metrics, particularly in diagnostic reasoning, follow-up recommendations, and explanatory clarity. These findings suggest that both models are probably better suited for clinical decision support tasks that require evidence-based reasoning, multi-step planning, and high interpretability.

However, several limitations of this study should be acknowledged. First, the evaluation was conducted within a single-center context and involved a limited number of expert raters, which may constrain the external validity and limit the generalizability of the findings across different clinical institutions, subspecialties, and practice patterns. Second, the current RAG-LLM framework was exclusively developed and assessed using textual inputs; critical multimodal data sources—such as radiological imaging, laboratory parameters, and histopathology reports—were not incorporated. This restricts the system's applicability in clinical scenarios where diagnostic accuracy relies heavily on image interpretation or integrated data synthesis. Third, although the proposed agents exhibited strong performance in static diagnostic reasoning and treatment planning, their real-world utility in broader clinical workflows—such as longitudinal patient follow-up, interdisciplinary decision-making, or integration into electronic health record (EHR) systems—remains unexplored. Lastly, the evaluation focused on the relative performance of different LLM agents within a unified RAG framework, without conducting a systematic, head-to-head comparison between AI-generated recommendations and the current standard of clinical care in spinal surgery (e.g., gold standards based on

authoritative guidelines, or actual diagnostic and treatment decisions made by multidisciplinary teams). Although five spinal surgeons provided ratings based on professional scales, this essentially constitutes an assessment of the "reasonableness" and "professionalism" of the AI outputs, rather than a direct validation of their alignment with best clinical practices. Therefore, our current results indicate differences in the capabilities of various AI models in generating structured, literature-supported spinal diagnosis and treatment recommendations, but we cannot yet assert that these recommendations are superior or equivalent to the decisions of experienced experts in terms of absolute clinical correctness. To address these limitations, future work should pursue multicenter validation with a larger and more diverse cohort of clinicians, extend the framework to support multimodal inputs using medical image encoders and structured data parsers, conduct prospective studies embedded in real-world clinical environments to evaluate usability, safety, and impact on patient outcomes, and design controlled experiments comparing AI suggestions with blinded expert committee resolutions or treatment pathways strictly adhering to guidelines, in order to more comprehensively evaluate their clinical effectiveness and safety.

Furthermore, the development and optimization of RAG-LLM agents are inherently sensitive to institutional practice patterns, which could lead to variability in system performance and design across different settings. For instance:

Knowledge Base Curation: The selection and weighting of source materials (e.g., prioritizing local institutional guidelines versus international consensus, or incorporating specific textbook chapters favored by a department) directly shape the evidence the system retrieves and, consequently, its recommendations.

Prompt Engineering and Output Expectations: The definition of an "appropriate" recommendation may vary. One institution may prefer highly detailed, conservative plans with extensive differentials, while another may value concise, action-oriented bullet points. This would necessitate tailored prompt templates and few-shot examples.

Therefore, while our framework provides a validated blueprint, its deployment in a new clinical environment would likely require a deliberate "localization" phase, involving collaboration with local experts to refine the knowledge base, calibrate prompts against local standards of care, and adapt to site-specific data workflows. This underscores the importance of considering not just the evaluation but also the adaptability of the development process itself when assessing the broader applicability of such clinical decision support tools.

5. Conclusion

This study presents a novel framework for developing RAG-based intelligent agents tailored to clinical decision support in spinal and spinal cord disorders. By integrating domain-specific medical knowledge with large language model inference, the proposed agents achieved superior performance in diagnostic accuracy, evidence traceability, and response interpretability. Furthermore, comprehensive case-based evaluations demonstrated the system's potential to assist clinicians in generating structured, explainable, and guideline-consistent recommendations. The modular architecture also supports future scalability toward multimodal data integration and real-time clinical deployment. Collectively, these findings demonstrate a functional framework and provide preliminary evidence supporting the clinical relevance of RAG-LLM framework for decision support. This work thereby contribute to the foundation for future research aimed at broader applications in specialty-specific, AI-driven healthcare.

CRediT authorship contribution statement

Yiren Li: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation. **Mingyu Lv:** Validation, Investigation. **Liang Cheng:** . **Yongxu Xie:** Writing – review & editing. **Qian Gu:** . **Haozheng He:** Data curation. **Zhuguang Chen:** Formal analysis. **Duopei Fang:** Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis. **Xiang Zhou:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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